

PoP-ECC: Robust and Flexible Error Correction against Multi-Bit Upsets in DNN Accelerators

Taewon Park[†], Saeid Gorgin[‡], Dongwhee Kim[†], Jaeho Shin[‡], Michael B. Sullivan[§], and Jungrae Kim[‡]

[†]Samsung Electronics, South Korea [‡]Sungkyunkwan University, South Korea [§]NVIDIA, USA

taeng1.park@samsung.com, gorgin81@skku.edu, dongwhee.kim@samsung.com, wogh533@g.skku.edu, misullivan@nvidia.com, dale40@skku.edu

Abstract—Deep Neural Networks (DNNs) in safety-critical systems require high reliability. Many systems deploy Error Correction Codes (ECCs) to protect DNNs from memory errors. However, continuous process scaling increases memory errors in severity and frequency, necessitating strong protection against Multi-Bit Upsets (MBUs). This paper proposes *Parities of Parities ECC (PoP-ECC)*, a novel two-tier memory protection scheme designed to provide robust, efficient, and flexible protection against MBUs. PoP-ECC generates *Virtual Parities (VPs)*, which are used to compute second-level parities called *Parities of Parities (PPs)*. This two-level ECC structure allows for dynamic error correction tailored to varying error patterns, ensuring system reliability with minimal memory overhead. Our evaluation demonstrates that PoP-ECC can tolerate significantly higher MBU ratios compared to state-of-the-art solutions, with negligible delay, area, and power overhead.

Index Terms—Reliability, ECC, SRAM, Deep Neural Network.

I. INTRODUCTION

With the proliferation of AI, Deep Neural Networks (DNNs) are widely used in safety-critical systems, including automotive [2], robotics [30], and aerospace [46] applications. DNN reliability is critical in such systems to ensure safety and prevent systems from failure. While DNNs are inherently resilient to small data noises, errors in high-bit positions can significantly impact their accuracy, posing a significant challenge to reliability.

One of the major concerns in DNN reliability is SRAM errors. A particle strike on SRAM can change the stored value, whereas combinational logic is less vulnerable to errors with temporal and logical masking [31]. Therefore, SRAM errors are already relatively common in modern systems [47], [48], and high-reliability systems protect their SRAMs with Error Correction Codes (ECCs). Moreover, DNN accelerators have a large amount of on-chip SRAM (e.g., 81% of chip area in Eyeriss [7]) to avoid high-energy off-chip memory accesses. With a large chip area and high error vulnerability, DNN requires strong SRAM protection for reliable operation.

Although most DNN reliability studies focus on protection against random bit errors, shrinking process technology demands strong protection against Multi-Bit Upsets (MBUs). With smaller feature sizes and lower supply voltage, the amount of charges to flip an SRAM cell (i.e., critical charge) keeps decreasing. As a result, many studies report that the probability of multiple adjacent errors is increasing [35], [32].

A common solution to DNN reliability is using ECC. However, the large SRAMs in DNN accelerators make the

conventional 12.5% ECC redundancy a significant overhead to overall chip size. To hide the storage overhead, some studies exploit the resiliency of DNN against small noises. They utilize aggressive quantization to reduce data size and make room for redundancy storage. For instance, Weight Nulling [37] saves one bit per weight and utilizes the bit as a parity to detect single-bit errors. If an error is detected, its erroneous weight is regarded as zero. Value-aware Parity Insertion ECC [29] replaces redundant under-utilized bits in a data block with ECC redundancy. These schemes can improve memory protection efficiency by re-purposing weight bits for redundancy and maintaining reliability levels in some models. However, they primarily target single-bit errors and provide weak protection against MBUs.

This paper proposes a novel memory protection scheme, called PoP-ECC, to enhance DNN reliability against MBUs. PoP-ECC starts by generating a *Virtual Parity (VP)* for each weight. Next, it creates a second level of redundancy, referred to as *Parities of Parities (PPs)*, based on these VPs. Instead of storing the VPs, PoP-ECC stores only the PPs alongside the weights. The decoding process re-generates VPs from the received data, restores corrupted VPs using PPs, and restores corrupted weights using VPs.

This two-level ECC provides both flexibility and efficiency in addressing MBU challenges. The VP generation can be tailored to handle various MBU patterns, with this paper focusing on Double Adjacent Errors (DAEs) due to their severity. Additionally, generating parities from VPs, which are smaller in size than their corresponding weights (e.g., 4 bits versus 6 or 8 bits), significantly reduces redundancy overhead compared to directly generating parities from weights.

Building on these strengths, this paper explores the integration of weight quantization with PoP-ECC to design an efficient SRAM protection scheme. The proposed method, Q+PoP, leverages the inherent robustness of DNNs to small data noises by applying quantization, thereby freeing up memory space for PoP-ECC redundancy. Through channel-wise quantization, Q+PoP minimizes accuracy degradation while ensuring sufficient space for PoP-ECC. This results in a highly efficient and reliable memory protection scheme tailored to modern DNN accelerators.

Our evaluation shows the efficiency and strength of Q+PoP. Although it quantizes weights to 6-bit, the channel-wise quantization leads to less than 0.5% in 2 out of 3 DNN models (1.02% on average). In terms of reliability, PoP-ECC

can maintain the accuracy without decreasing by $> 0.1\%$ percentage point even if the bit error ratio (BER) increases up to 3×10^{-3} ; the ratio is up to $3.16 \times$ higher than state-of-the-art schemes. More importantly, it can tolerate up to $31.62 \times$ higher DAE ratio than the state-of-the-art schemes. By providing an order of magnitude stronger protection against MBUs, PoP-ECC can improve DNN reliability significantly, as overall reliability is dominated by the weakest protection.

The major contributions of this paper are as follows:

- This paper introduces a novel ECC scheme called *Parities-of-Parities ECC (PoP-ECC)*, which protects virtual data instead of raw data. PoP-ECC can customize error correction to target MBU patterns and reduces redundancy, making it highly efficient for modern SRAM reliability challenges.
- Our evaluation demonstrates that PoP-ECC tolerates up to $31.62 \times$ higher DAE ratios than state-of-the-art schemes, with less than 0.1 percentage points accuracy degradation compared to error-free cases, ensuring strong protection with minimal impact on performance.
- The proposed *Q+PoP* integrates channel-wise quantization with PoP-ECC to enhance the reliability of DNN inference. This approach provides robust protection against MBUs while preserving the SRAM footprint, balancing efficiency and reliability.

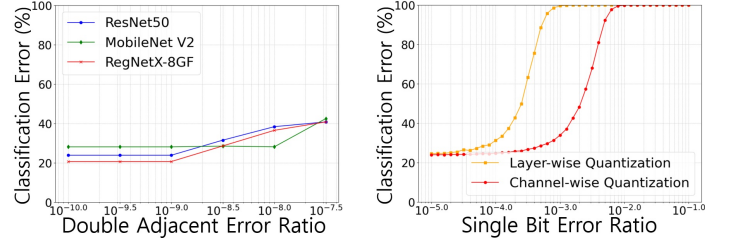
II. BACKGROUND AND MOTIVATION

This section introduces the impacts of SRAM errors and quantization on DNN reliability, which motivated this study.

A. DNNs and SRAM Errors

DNN processors commonly equip a large amount of SRAMs. The primary design target of the processors is energy efficiency, yet external DRAM access consumes more than an order of magnitude higher energy than on-chip SRAMs [22]. Accordingly, many DNN processors size up on-chip SRAM buffers and optimize reuse patterns to keep DRAM accesses to minimum [7], [28], [23], [18], [16], [50], [14], [19]. Eyeriss[7], for instance, utilizes 81% of its core chip area (i.e., the area without I/O pads) for its global buffer, scratchpad memory, and FIFOs. At the extreme, many DNN accelerators do not feature an external DRAM interface [4], [8], [15], [13], [27], [17]. For example, Cerebras WSE2 has 40GB on-chip SRAM over a wafer-scale chip [4].

Meanwhile, SRAM is more error-vulnerable than logic cells, which have temporal masking and logical masking [31]. Therefore, functional safety standards for autonomous vehicles and industrial robotics also require very strong SRAM reliability against soft errors (ISO 26262 [25] and IEC 61508 [24]). In addition, computers in real-world systems are subject to elevated temperatures, and they often operate at low voltages to reduce energy usage; these factors can increase the soft error rate in SRAMs [41]. Combined with the large chip area, the higher vulnerability makes SRAM a dominant threat to DNN processing.



(a) DNN reliability against double adjacent errors

(b) Reliability of layer- vs. channel-wise quantization

Fig. 1: DNN top-1 accuracy degradation with memory errors.

Although most DNN SRAM reliability studies focus on single-bit errors, protection against multi-bit errors is increasingly important. Shrinking process technology and decreasing supply voltage reduces the amount of charges to upset a memory cell so that a high-energy particle strike can affect multiple nearby cells, a symptom known as *Multi-cell Upset (MCU)*. In a recent 5nm FinFET technology, 25% of alpha particle events and 24% of terrestrial neutron events are MCUs [36]. And the ratio increases to 78% and 93%, as neutrons' energy increases to 21 MeV and 85 MeV, respectively. Other experiments present similar or higher MCU ratios [33], [45], [35], [9], [34]. An MCU can affect multiple bits in a word, or spread errors across different words; the former is called as *Multi-bit Upset (MBU)* [1].

The cost of ignoring MBUs in DNN protection can be severe. Errors can corrupt the upper bits of DNN data to change the value significantly. To evaluate the cost, we conducted a DNN accuracy evaluation with DAE injections. We assume all single errors are correctable and inject DAEs only with a uniform random distribution and varying ratios. Errors in activation or weight can affect DNN reliability. However, prior work reports that weights are less error-tolerant than activations because new inputs continuously update activations while weights are reused many times [31], [39]. Therefore, we inject DAEs exclusively into the weights. The result (Fig. 1a) shows that the accuracy is significantly affected by DAEs at a relatively modest ratio of 10^{-8} .

B. DNNs and Quantization

DNN inference is known to be resilient to small noises. Quantization [11] exploits this resilience and converts DNN data in real values into low bit-width numbers, significantly reducing memory usage and computational costs. A popular quantization scheme is the uniform quantization with scaling [26]. It uses a scale parameter to divide a given range of real values into integer values and a zero-point parameter to specify the offset. These parameters are extracted from the data set to avoid mapping to unused values.

There are two ways of applying uniform quantization: layer-wise quantization applies to all weights of each layer, whereas channel-wise quantization applies to all weights of each filter. Channel-wise quantization has different scales and zero-point parameters for each filter and can provide better data resolution and DNN accuracy. Table I compares DNN

TABLE I: Accuracy degradation of quantization methods

Model	Baseline (FP32)	8bit		6bit	
		Channel	Layer	Channel	Layer
ResNet50	76.13%	(-0.03%)	(-0.12%)	(-0.41%)	(-3.00%)
MobileNet V2	71.88%	(-0.79%)	(-1.66%)	(-2.47%)	(-14.97%)
RegNet-X-8GF	79.34%	(+0.04%)	(-0.05%)	(-0.17%)	(-2.00%)
Average		(-0.26%)	(-0.61%)	(-1.02%)	(-6.66%)

accuracy with different quantization schemes. It shows that channel-wise quantization can provide better accuracy with smaller quantization errors, and the accuracy gap increases with more aggressive quantization (i.e., 6-bit quantization).

Figure 1b shows the ResNet50 [21] classification errors with varying bit error ratios. In this experiment, we assume quantization parameters are stored reliably and inject errors only into quantized weights. The result shows that channel-wise quantization is more robust than layer-wise quantization because it has a smaller range for quantized weights, and an error cannot change the value outside the range. Therefore, Q+PoP utilizes channel-wise quantization to improve accuracy and reliability.

III. PRIOR WORKS

A. Error Correction Codes

Error Correction Codes (ECCs) can correct errors to enhance memory reliability. Most memory subsystems utilize linear block codes with simpler decoding to reduce read latency. Linear block codes can be classified based on their symbol size: binary ECC and non-binary ECC. Binary ECC codes correct errors on a bit basis. The most well-known binary ECC is Hamming code [20]. It can correct the one-bit error (i.e., Single Error Correction (SEC)) in n -bit codewords using $\lceil \log_2(n+1) \rceil$ redundant bits. Single Error Correction - Double Error Detection (SEC-DED) codes extend SEC by adding one-bit parity. Binary BCH codes [3] are other popular binary ECC schemes. BCH codes can efficiently correct random multi-bit errors. For n -bit codewords, BCH codes require $t \times \lceil \log_2(n+1) \rceil$ redundant bits to correct t -bit errors.

Non-binary ECC codes can correct errors with a symbol basis. Among non-binary symbol ECCs, Reed Solomon (RS) codes are popular in memory ECC [40]. RS codes can correct t symbol errors using $2t$ redundant symbols if the codeword is no bigger than $2^m - 1$ symbols, where m is the symbol size. Although RS codes can correct multiple erroneous bits within a symbol within a single correction, which is good at burst errors, they are relatively inefficient in correcting randomly distributed bit errors.

Some prior work proposed applying ECC in multiple levels. Methods such as LOT-ECC [49], RATT-ECC [5], and Configurable-ECC [6] employ two-tiered ECC schemes for DRAM. In these approaches, errors that cannot be corrected by the first tier are addressed by the second tier, which distributes redundancy across multiple tier-1 blocks to minimize overhead.

B. DNN Reliability Enhancement via ECC

Many studies proposed weight protection schemes to enhance DNN reliability. The simplest way of protecting the

weights is applying ECC to the weights and storing the redundant check bits in extra storage. However, SRAMs occupy a large area in DNN processors, and reducing the protection cost is one of the main challenges for DNN protection. Quantization is the most effective technique for eliminating protection storage costs, leveraging the inherent resilience of DNNs.

Weight Nulling [37] replaces the least significant bit of a half-precision floating-point weight with one-bit parity to detect errors. A weight with a detected error is regarded as zero and does not participate in computation. Therefore, it cannot protect against a high error ratio as more errors nullify more weights, potentially leading to degraded accuracy. Moreover, multi-bit errors within a weight can escape the parity check, leading to uncorrected weight corruption.

In-place Zero-space ECC [12] utilizes that most weights have small absolute values. It first retrains DNN models with much smaller weights near zero, allowing efficient representation with 7-bit quantized values. The remaining non-informative 1-bit is used for redundancy. Then, eight 7-bit quantized weights are grouped and encoded with a (64, 57) SEC-DED code to form a 64-bit block. This can provide correction for single-bit errors, while many soft errors are multi-bit upsets (MBUs). Additionally, retraining requires significant computation and may impact accuracy.

Value-aware Parity Insertion ECC [29] is one of the strongest DNN protection schemes. Similar to [12], it leverages a narrow weight distribution for aggressive quantization and redundancy storage. Weights are represented using a custom 1-bit sign and 7-bit magnitude format, assuming that most weights fall below 0.5. It allows the two most significant bits to be used for storing parity. If weights fall outside the range of $[-0.5, 0.5]$, the two least significant bits are truncated and used for parity. This scheme provides Double Error Correction (DEC) to 64-bit blocks, improving error tolerance. However, it may reduce DNN accuracy, particularly when weights fall outside the expected range or due to the loss of weight values. Moreover, its DEC capabilities may be insufficient in a higher error ratio of MBUs.

IV. PARITIES OF PARITIES ECC (PoP-ECC)

PoP-ECC is a strong and flexible scheme to protect data from severe memory errors. In designing PoP-ECC, we aim to protect data from memory MBUs with minimal costs. PoP-ECC is a two-level ECC; the first level encoding extracts *Virtual Parities (VPs)* from each value, and the second level protects VPs using *Parities of Parities (PPs)*. VPs are used for the second-level parity generation only and are not stored in memory. One of the key differences of PoP-ECC is that it protects virtual data, called VP, whereas prior works protect raw data. Extracting a VP from raw data is flexible and can be customized to the target MBUs and applications.

PoP-ECC is a general method and operates efficiently across various precisions and sizes. Without loss of generality, in Q+PoP, we use a lower-precision representation for

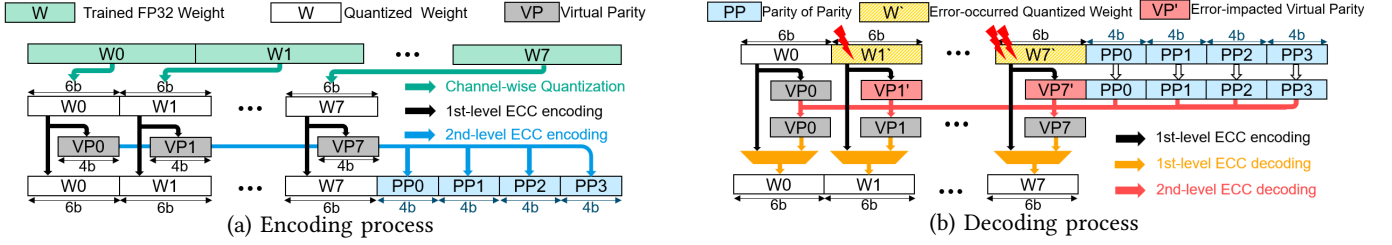


Fig. 2: An overview of Q+PoP

clarity and consistency with prior works, reducing from 8-bit to 6-bit, to minimize protection costs and allocate space for PoP-ECC redundancy. Using 6-bit precision, 16 bits are saved per 64-bit block, allowing Q+PoP to utilize the saved space for storing redundant protection bits. We choose VP as a 4-bit parity to correct all single-/double-bit errors on the upper 5 bits of a quantized value; we do not protect the least significant bit because errors in the bit position are not critical to accuracy in our evaluation.

A. Encoding Process

Figure 2a illustrates the encoding process of PoP-ECC within the Q+PoP framework. Prior to encoding, data precision is reduced to 6-bit integers using channel-wise quantization. The encoding process in PoP-ECC operates in two stages. In the first stage, a 4-bit Virtual Parity (VP) is generated from the 6-bit quantized data. This stage allows for flexible customization of the ECC encoding scheme to address varying MBU severity and patterns. The generated 4-bit VP is designed to correct all single- and double-bit errors in the upper 5 bits of the quantized weight. Notably, the lowest bit is left unprotected, as experiments reveal that errors in this position have negligible impact on accuracy.

The upper 5 bits can have five single-bit errors and ten ($=_5C_2$) double-bit errors, and the 4-bit VP uses its 15 non-zero syndromes to correct them. The encoding utilizes the 5×9 generator matrix to generate a 4-bit VP from the upper 5-bits of a quantized weight. It generates unique syndromes for each single- and double-bit error on the weight. The parity check matrix (4×9) has identical columns in the data and parity parts, which is unusual. However, the virtually stored parities are error-free and cannot generate a syndrome the same as data errors. It can use a table (optimized during synthesis) to locate errors from the syndromes and restore the weight. VP generation increases the encoding and decoding latency slightly with only one level of XOR gates.

In the second stage, PoP-ECC utilizes RS code to protect VPs on a symbol basis. Because each VP is 4-bit, it applies 4-bit RS encoding to the eight VPs to generate four redundant symbols (i.e., PPs) and stores the data and PPs. It is worth mentioning that PoP-ECC differs from concatenated codes [10] in that its second-level ECC encodes VPs only and does not directly protect data. Moreover, it does not store the VP. The total block size remains 64-bit ($8 \times \{6\text{-bit quantized weights}\} + 4 \times \{4\text{-bit PPs}\}$), whereas the symbol count is 12 (8 VPs and 4 PPs).

B. Decoding Process

Figure 2b illustrates the decoding process of PoP-ECC, with some errors on the stored block. In the example, protected data, W1 and W7, have a single- and double-error, respectively. On receiving the erroneous block, PoP-ECC first regenerates VPs using the same generator matrix as the encoding (i.e., first-level ECC of PoP-ECC). The errors in this example belong to the upper 5 bits of weights, and some regenerated VPs (VP1 and VP7) differ from the originals in the encoding. The second-level ECC applies the RS decoding on VPs and PPs, and it can restore the two corrupted VPs using two symbol corrections. After restoring the VPs, PoP-ECC utilizes the VPs to restore the weights; VP1 can correct the single error on W1, while VP7 can correct the double error on W7. The VP regeneration and the VP-based correction may affect the decoding latency, yet our evaluation shows they do not increase the latency significantly ($\leq 0.18ns$ in UMC 28nm process, Section V-D).

Without an error, the regenerated VPs will be the same as the original VPs, leaving no correction by VPs and PPs. Similarly, if errors corrupt the lowest bits of data only, the regenerated VPs will be the same as the original VP (the bit does not participate in VP generation), and the errors will be neither detected nor corrected. However, our experiments show small noises from these errors do not significantly affect accuracy. Compared to the selective error correction on quantized weights, PoP-ECC can correct any number of errors on a PP using the symbol-based correction unless the number of erroneous weights and erroneous PPs exceed its correction capability (i.e., 2).

In addition to stronger protection, PoP-ECC can provide flexible protection. If some error patterns are dominant (e.g., DBU) or some bits are more/less critical (e.g., the least significant bit is non-critical), it can tailor the VP generation scheme. Moreover, PoP-ECC can control the number of PPs (e.g., 2 PPs for single symbol correction and 4 PPs for double symbol correction) to deal with different error rates.

V. EVALUATION

We evaluate the accuracy and reliability of Q+PoP, which integrates PoP-ECC, and compare its performance to state-of-the-art schemes. The results demonstrate that Q+PoP, leveraging channel-wise quantization, achieves superior base accuracy, ensuring minimal accuracy loss in error-free scenarios while enhancing reliability. In terms of fault tolerance, Q+PoP can withstand significantly higher ratios of single-bit

TABLE II: Accuracy degradation of the protection schemes

Model	Baseline (FP32)	Weight Nulling (FP16)	VAPI (8b Sign/Mag. Q.)	Q+PoP (6b Ch-wise Q.)
ResNet50	76.13%	(0.00%)	(-4.25%)	(-0.41%)
MobileNet V2	71.88%	(-0.01%)	(-16.07%)	(-2.47%)
RegNet-X-8GF	79.34%	(+0.01%)	(-2.01%)	(-0.17%)
Average		(0.00%)	(-7.44%)	(-1.02%)

errors and DAEs with stronger correction and error-robust quantization. Despite these advancements in accuracy and reliability, Q+PoP incurs minimal hardware overhead, thanks to its use of small symbols.

A. Methodology

We implemented Q+PoP and two state-of-the-art DNN protection schemes (Weight Nulling and Value-aware Parity Insertion (VAPI)) in the PyTorch framework. We do not include In-place Zero-space ECC because it provides weak protection (SEC-DED) and requires retraining the model. All evaluated schemes have 64-bit block sizes. Weight Nulling has four FP16 weights in the block and replaces the least significant bits with one-bit parity to detect errors in the weight. VAPI has eight weights in the 1-bit sign and 7-bit magnitude format and utilizes two magnitude bits to store ECC redundancy.

We use three representative DNN models (ResNet50 [21], MobileNet V2 [43], and RegNetX-8GF [38]) with Torchvision pretrained weights. The input dataset is ILSVRC-2012 [42], and we measure top-1 classification accuracy over 50,000 validation images. To measure the accuracy, we run the models with quantized weights in FP32 format. We map quantized weights to their closest FP32 values. This provides a common execution environment across different data types (i.e., FP16 in Weight Nulling, sign/magnitude in VAPI, and INT6 in Q+PoP), while the high precision (24-bit significant in FP32) minimizes conversion losses down to 2^{-24} of the original values and minimizes its impact on accuracy.

B. Accuracy

We first run the DNN models without errors and measure the inference accuracy. All evaluated schemes apply aggressive quantization to create room for redundancy. To quantify the accuracy cost of the quantization, we measure DNN accuracy without injecting errors.

Table II compares the inference accuracy without errors. Weight Nulling provides almost the same accuracy as the pretrained weights because replacing the least significant bit of FP16 with one-bit parity does not affect accuracy significantly. VAPI sacrifices accuracy significantly (up to 16.07% in MobileNet) in our evaluation.

Meanwhile, Q+PoP shows an average of 1.02% degradation. Despite aggressive 6-bit quantization, its channel-wise quantization can flexibly deal with diverse weight scales. 2 out of 3 models have $< 0.5\%$ degradation, while MobileNet V2 shows the biggest degradation of 2.47%. MobileNet is a depthwise-separable CNN, reported as more difficult to quantize than other CNNs [51].

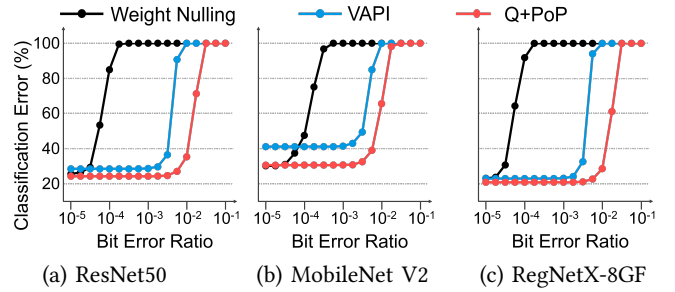


Fig. 3: DNN reliability comparison against varying single-bit error ratios.

C. Reliability

Then we run the DNN models with errors to evaluate reliability. We inject two types of errors—Single Error (SE) and DAE, representing Single-Bit Upsets and MBUs, respectively. Errors are applied only to weight blocks, as weights are crucial to DNN reliability [39], [29]. Each target block is randomly selected, formatted to the protection scheme’s layout, encoded with ECC, and parity bits are stored in designated positions. Errors are then injected at random positions within the block, including parity bits, simulating real-world scenarios.

After injecting errors, the ECC decoding process is applied to the block. Successful correction restores the quantized weight, while miscorrection worsens the error by introducing new ones. Detected but uncorrectable errors leave the weight unchanged, except in cases of Weight Nulling, which nullifies it. Decoded weights are converted back to FP32, with uncorrected weight errors altering values based on the scheme’s format. Errors in parity are ignored to mimic real scenarios. The process is repeated 50 times with various error injections, and the average classification error is calculated.

1) *Reliability against Single-Errors*: Figure 3 presents the results of the SE injection experiments. We sweep the Bit Error Ratio (BER) from 10^{-5} to 10^{-1} and measure the inference accuracy.

For lower BERs (e.g., $< 10^{-5}$), the accuracy results are almost identical to those without errors in Table II. At these relatively low ratios, Q+PoP can provide significantly higher accuracy than VAPI (-1.02% vs. -7.44% accuracy loss) with its channel-wise quantization. Weight Nulling provides almost no accuracy loss. However, it is not energy-efficient due to its doubled memory footprint (16-bit vs. 8-bit per weight) and energy-expensive floating-point operations (FP16).

The accuracy of Weight Nulling decreases quickly as the ratio approaches 10^{-4} . At the ratio, 1.6% of FP16 weights have single-bit errors and are nullified. Moreover, the accuracy quickly approaches 0% as the ratio increases. Increasing errors not only nullifies more weights with single errors but also generates undetectable errors from double errors on a weight. The one-bit parity cannot detect an even number of errors and degrades accuracy significantly at a high BER or a relatively low DAE ratio (Section V-C2).

Q+PoP can tolerate the highest BER of up to near 3×10^{-3} .

TABLE III: The maximum BERs which do not reduce DNN accuracy by 0.1 percentage point than error-free cases.

Model	VAPI	Q+PoP	Ratio
ResNet50	$10^{-3.2}$	$10^{-2.8}$	$2.51\times$
MobileNet V2	$10^{-3.2}$	$10^{-3.1}$	$1.26\times$
RegNet-X-8GF	$10^{-3.2}$	$10^{-2.7}$	$3.16\times$

TABLE IV: The maximum DAE ratios which do not reduce DNN accuracy by 0.1 percentage point than error-free cases.

Model	VAPI	Q+PoP	Ratio
ResNet50	$10^{-4.2}$	$10^{-3.1}$	$12.59\times$
MobileNet V2	$10^{-4.3}$	$10^{-3.4}$	$7.94\times$
RegNet-X-8GF	$10^{-4.5}$	$10^{-3.0}$	$31.62\times$

It can correct not only single and double errors but also parts of triple and quadruple errors as long as they belong to two quantized weights, and each weight has up to 2 errors. The stronger correction allows Q+PoP to tolerate higher error ratios than VAPI with DEC, and the channel-size quantization minimizes accuracy degradation from uncorrectable errors. To quantify the improved reliability, we checked error ratios at which the accuracy degraded by 0.1 percentage point than no error cases. Table III shows Q+PoP can tolerate up to $3.16\times$ higher BERs than VAPI.

2) *Reliability against Double-Adjacent Errors:* Figure 4 presents the results of DAE injection experiments. Like the SE results, Q+PoP is significantly more robust than Weight Nulling and VAPI. Weight Nulling cannot detect double-bit errors within a weight and provides protection as bad as no protection against DAEs (almost 0% accuracy at 10^{-5} and higher ratios).

VAPI can correct one DAE using its DEC BCH codes to tolerate up to 8×10^{-5} . However, at 10^{-3} , the chances of having two DAEs within a 64-bit block increase to 6.3%, and the accuracy decreases to near 0%. Using the same amount of redundancy, PoP-ECC can tolerate up to 10^{-3} by correcting up to two DAEs within a 64-bit block. Combined with better base accuracy, the less accuracy degradation make Q+PoP a accurate and strong solution for reliable DNN processing. Table IV compares the maximum DAE ratio, which does not reduce DNN accuracy by 0.1 percentage point than error-free cases. It shows Q+PoP can tolerate up to $31.62\times$ higher DAE ratios than VAPI.

Overall, Weight Nulling provides the highest baseline accuracy, yet the accuracy is based on double memory footprint (16-bit) and energy-expensive floating point operations. VAPI can tolerate relatively-high error ratios (e.g., $10^{-3.2}$ BER and $10^{-4.2}$ DAE ratio), yet the baseline accuracy can be poor due to its inflexible quantization (e.g., up to 16.07% accuracy loss, Table II). Q+PoP provides both high accuracy and reliability with 1.02% baseline accuracy degradation, which is comparable to INT8 quantization schemes. And it maintains the fair accuracy up to very high error ratios.

D. Hardware Costs

To evaluate the hardware overheads of PoP-ECC, we implemented its encoder and decoder in Verilog. The decoder

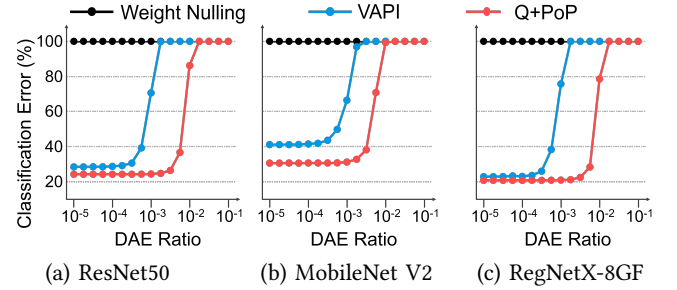


Fig. 4: DNN reliability comparison against varying Double-Adjacent Error (DAE) ratios.

follows an algorithm in [44] for faster decoding. We synthesize the Verilog models using Synopsys Design Compiler and UMC 28nm SVT library.

The encoding takes $0.65ns$, which is small enough to be implemented in 1 cycle in modern technology nodes. At a deeper look, VP generation requires only one level of 2-input XOR gates and takes $0.22ns$. PP generation mostly comprises XOR trees and takes the remaining $0.43ns$. The decoding takes $1.56ns$, which can be split into error detection and error correction. The detection part is similar to the encoding and takes $0.8ns$. We expect this will not increase the read latency than SEC-DED. The correction part takes an additional $0.8ns$, which can be skipped in no error cases.

The area overhead is $92\mu m^2$ (0.3K NAND2 gates) for the encoder and $1760\mu m^2$ (6K NAND2 gates) for the decoder. Most of the increase in the decoder comes from the RS decoder. The power consumption of the encoder and decoder are 0.08mW and 1.2mW, respectively.

VI. CONCLUSION

This paper introduces PoP-ECC, a novel two-level ECC scheme designed to address growing reliability challenges in SRAMs. By leveraging Virtual Parities and Parities of Parities, PoP-ECC provides a robust and flexible solution for mitigating MBUs. Building on the strengths of PoP-ECC and channel-wise quantization, we also propose Q+PoP, a robust and efficient DNN protection scheme. Our evaluation demonstrates that Q+PoP achieves better base accuracy and can tolerate up to $31.62\times$ higher DAE ratios compared to state-of-the-art schemes, all with negligible overhead in delay, area, and power.

ACKNOWLEDGMENT

This work was partly supported by Institute of Information & communications Technology Planning & Evaluation (IITP) grant funded by the Korea government (MSIT) (RS-2019-II190421, Artificial Intelligence Graduate School Program (Sungkyunkwan University), 50%), and the Technology Innovation Program (or Industrial Strategic Technology Development Program-Public-Private Joint Investment Advanced Semiconductor Talent Development Project) (RS-2023-00237136, Development of CXL/DDR5-based memory subsystem for AI accelerators, 50%).

REFERENCES

- [1] *Measurement and Reporting of Alpha Particle and Terrestrial Cosmic Ray-Induced Soft Errors in Semiconductor Devices*, JESD89B, Joint Electronic Device Engineering Council, 2006.
- [2] M. Borg *et al.*, "Safely Entering the Deep: A Review of Verification and Validation for Machine Learning and a Challenge Elicitation in the Automotive Industry," *Journal of Automotive Software Engineering*, vol. 1, no. 1, 2019.
- [3] R. C. Bose and D. K. Ray-Chaudhuri, "On a class of error correcting binary group codes," *Information and control*, vol. 3, no. 1, 1960.
- [4] Cerebras Systems, "The Future of AI is Here," <https://www.cerebras.net/product-chip/>, 2023.
- [5] H.-M. Chen *et al.*, "RATT-ECC: Rate Adaptive Two-Tiered Error Correction Codes for Reliable 3D Die-Stacked Memory," *ACM Transactions on Architecture and Code Optimization (TACO)*, vol. 13, no. 3, 2016.
- [6] H.-M. Chen, S.-Y. Lee, T. Mudge, C.-J. Wu, and C. Chakrabarti, "Configurable-ecc: Architecting a flexible ecc scheme to support different sized accesses in high bandwidth memory systems," *IEEE Transactions on Computers*, vol. 68, no. 5, 2019.
- [7] Y.-H. Chen, T. Krishna, J. S. Emer, and V. Sze, "Eyeriss: An Energy-Efficient Reconfigurable Accelerator for Deep Convolutional Neural Networks," *IEEE Journal of Solid-State Circuits*, vol. 52, no. 1, 2017.
- [8] M. Denler, "Hailo Illuminates Low-Power AI Chip," The Linley Group Microprocessor Report from June 24, 2019.
- [9] V.-P. Fang and A. Oates, "Cell Level Soft Error Rate Simulations on Planar and FinFET Processes," in *Proceedings of the CPMT SER Workshop*, 2014.
- [10] G. D. Forney, "Concatenated Codes," Massachusetts Institute of Technology, Tech. Rep. 440, 1965.
- [11] A. Gholami, S. Kim, Z. Dong, Z. Yao, M. W. Mahoney, and K. Keutzer, "A survey of quantization methods for efficient neural network inference," 2021. [Online]. Available: <https://arxiv.org/abs/2103.13630>
- [12] H. Guan *et al.*, "In-place zero-space memory protection for CNN," in *Proceedings of the 33rd International Conference on Neural Information Processing Systems (NIPS)*, 2019.
- [13] L. Gwennap, "Graphcore Makes Big AI Splash," 2018, The Linley Group Microprocessor Report from September 17, 2018.
- [14] —, "Habana Wins Cigar for AI Inference," 2018, The Linley Group Microprocessor Report from February 18, 2018.
- [15] —, "Alibaba Takes AI Performance Lead," 2019, The Linley Group Microprocessor Report from November 25, 2019.
- [16] —, "Graphcore Doubles IPU Performance," 2020, The Linley Group Microprocessor Report from August 3, 2020.
- [17] —, "Groq Rocks Neural Networks," 2020, The Linley Group Microprocessor Report from January 6, 2020.
- [18] —, "Tenstorrent Scales AI Performance," 2020, The Linley Group Microprocessor Report from April 13, 2020.
- [19] —, "Quadric's Flexible Edge Accelerator," 2021, The Linley Group Microprocessor Report from July 5, 2021.
- [20] R. W. Hamming, "Error detecting and error correcting codes," *The Bell system technical journal*, vol. 29, no. 2, 1950.
- [21] K. He *et al.*, "Deep residual learning for image recognition," in *Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR)*, 2015.
- [22] M. Horowitz, "Energy Table for 45nm Process," in *Stanford VLSI Wiki*, 2014.
- [23] S. Hosseiniinorbin *et al.*, "Exploring edge tpu for deep feed-forward neural networks," *Internet of Things*, vol. 22, 2023.
- [24] International Electrotechnical Commission, "Functional Safety of Electrical/Electronic/Programmable Electronic Safety-related Systems," <https://webstore.iec.ch/publication/22273>, 2010.
- [25] International Organization for Standardization, "ISO 26262:2018 Road Vehicles—Functional Safety," <https://www.iso.org/standard/68383.html>, 2018.
- [26] B. Jacob *et al.*, "Quantization and training of neural networks for efficient integer-arithmetic-only inference," in *Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR)*, 2018.
- [27] A. Jani, "Mythic Unveils Analog-AI Product," 2020, The Linley Group Microprocessor Report from November 30, 2020.
- [28] N. P. Jouppi *et al.*, "In-datacenter performance analysis of a tensor processing unit," in *Proceedings of the 44th Annual International Symposium on Computer Architecture (ISCA)*, 2017.
- [29] S.-S. Lee and J.-S. Yang, "Value-Aware Parity Insertion ECC for Fault-Tolerant Deep Neural Network," in *Proceedings of the Conference on Design, Automation, and Test in Europe (DATE)*, 2022.
- [30] S. Levine *et al.*, "End-to-End Training of Deep Visuomotor Policies," *The Journal of Machine Learning Research*, vol. 17, no. 1, 2016.
- [31] G. Li *et al.*, "Understanding Error Propagation in Deep Learning Neural Network (DNN) Accelerators and Applications," in *Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis (SC)*, 2017.
- [32] R. K. Maity, J. Samanta, and J. Bhaumik, "Single and double-adjacent error correcting code (SDECC) with lower design overheads and mis-correction rate for SRAMs," *Microsystem Technologies*, 2023.
- [33] D. G. Mavis *et al.*, "Multiple bit upsets and error mitigation in ultra-deep submicron srams," *IEEE Transactions on Nuclear Science*, vol. 55, no. 6, 2008.
- [34] B. Narasimham *et al.*, "Scaling Trends and Bias Dependence of the Soft Error Rate of 16 nm and 7 nm FinFET SRAMs," in *Proceedings of the International Reliability Physics Symposium (IRPS)*, 2018.
- [35] A. Neale and M. Sachdev, "Neutron Radiation Induced Soft Error Rates for an Adjacent-ECC Protected SRAM in 28nm CMOS," *IEEE Transactions on Nuclear Science*, vol. 63, no. 3, 2016.
- [36] N. J. Pieper, Y. Xiong, A. Feeley, J. Pasternak, N. Dodds, D. R. Ball, and B. L. Bhuvu, "Study of multi-cell upsets in sram at a 5-nm bulk finfet node," *IEEE Transactions on Nuclear Science*, 2023.
- [37] M. Qin *et al.*, "Robustness of Neural Networks against Storage Media Errors," 2017. [Online]. Available: <https://arxiv.org/abs/1709.06173>
- [38] I. Radosavovic *et al.*, "Designing network design spaces," in *Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR)*, 2020.
- [39] B. Reagen *et al.*, "Ares: A Framework for Quantifying the Resilience of Deep Neural Networks," in *Proceedings of the Design Automation Conference (DAC)*, 2018.
- [40] I. S. Reed and G. Solomon, "Polynomial codes over certain finite fields," *Journal of the society for industrial and applied mathematics*, vol. 8, no. 2, 1960.
- [41] M. Riera, R. Canal, J. Abella, and A. Gonzalez, "A Detailed Methodology to Compute Soft Error Rates in Advanced Technologies," in *2016 Design, Automation Test in Europe Conference Exhibition (DATE)*, 2016.
- [42] O. Russakovsky *et al.*, "Imagenet Large Scale Visual Recognition Challenge," *International Journal of Computer Vision (IJCV)*, vol. 115, no. 3, 2015.
- [43] M. Sandler *et al.*, "Mobilenetv2: Inverted residuals and linear bottlenecks," in *Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR)*, 2018.
- [44] D. V. Sarwate and N. R. Shanbhag, "High-speed architectures for reed-solomon decoders," *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 9, no. 5, 2001.
- [45] N. Seifert, B. Gill, S. Jahinuzzaman, J. Basile, V. Ambrose, Q. Shi, R. Allmon, and A. Bramnik, "Soft Error Susceptibilities of 22 nm Tri-Gate Devices," *IEEE Transactions on Nuclear Science*, vol. 59, no. 6, 2012.
- [46] M. Shirobokov, S. Trofimov, and M. Ovchinnikov, "Survey of machine learning techniques in spacecraft control design," *Acta Astronautica*, vol. 186, 2021.
- [47] V. Sridharan *et al.*, "Feng shui of supercomputer memory positional effects in dram and sram faults," in *Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis (SC)*, 2013.
- [48] —, "Memory errors in modern systems: The good, the bad, and the ugly," *ACM SIGARCH Computer Architecture News*, vol. 43, no. 1, 2015.
- [49] A. N. Udiipi *et al.*, "LOT-ECC: Localized and Tiered Reliability Mechanisms for Commodity Memory Systems," in *Proceedings of the International Symposium on Computer Architecture (ISCA)*, 2012.
- [50] B. Wheeler, "Flex Logix Shows First AI Chip," The Linley Group Microprocessor Report from November 16, 2020.
- [51] S. Yun and A. Wong, "Do All MobileNets Quantize Poorly? Gaining Insights into the Effect of Quantization on Depthwise Separable Convolutional Networks Through the Eyes of Multi-scale Distributional Dynamics," in *Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR)*, 2021.